

Bansilal Ramnath Agarwal Charitable Trust's
Vishwakarma Institute of Information Technology, Pune-48
(An Autonomous Institute affiliated to Savitribai Phule Pune University)



Curriculum for
Third Year B. Tech.
Computer Science and Engineering
(Artificial Intelligence)
2020R1 Pattern

Department of Computer Science and Engineering
(Artificial Intelligence)



Vishwakarma Institute of Information Technology, Pune-48
(An Autonomous Institute affiliated to Savitribai Phule Pune University)
Department of Computer Science and Engineering (Artificial Intelligence)

Vision:

“Excellence in Computer Science and Engineering providing AI-driven solutions with holistic development.”

Mission:

- To equip aspiring engineers for industry, academia, research, and entrepreneurship by offering excellent education in emerging AI techniques.
- To impart value-added technical and research-oriented education by inculcating life skills with an emphasis on human values.
- To educate and empower Learners with professional integrity for research and higher studies.

Program Specific Outcomes (PSOs):

At the end of the program, students should be able to

PSO1: Apply problem solving skills through programming techniques for addressing real life problems using suitable AI Techniques to provide innovative computing solutions.

PSO2: Solve industrial and societal problems using innovative AI techniques with ethical considerations.

Programme Educational Objectives (PEOs):

1. Develop graduates who possess applied knowledge and skills in artificial intelligence, enabling them to effectively analyze complex problems and devise innovative solutions in diverse domains
2. Provide students with holistic and multidisciplinary expertise in computing, with a focus on artificial intelligence, to foster innovation throughout their careers.
3. Impart the importance of good ethical practices, professional conduct, responsible team leadership, and societal impact among students in the AI domain.

Program Outcomes (POs)

At the end of program, students should be able to:

1. **Engineering knowledge:** Apply the knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.



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2. **Problem analysis:** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences.
3. **Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety and the cultural, social and environmental considerations.
4. **Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
5. **Modern tool usage:** Create, select and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.
6. **The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
7. **Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts and demonstrate the knowledge of, and need for sustainable development.
8. **Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
9. **Individual and team work:** Function effectively as an individual and as a member or leader in diverse teams and in multidisciplinary settings.
10. **Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations and give and receive clear instructions.
11. **Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
12. **Life-long learning:** Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.



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Third Year B.Tech Pattern 2020R1 Syllabus (Year 2024-25)



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T.Y. B.Tech. Computer Science and Engineering (Artificial Intelligence)
SEMESTER V (PATTERN 2020R1)

CourseCode	Course Name	Teaching Scheme (Hrs/Week)				Assessment Scheme (100-mark scale)										Credits
		Course Type	Theory	Lab	Tut	ISA						ESA			Total	
						HA	TW	SCE	PPT	GD	CIE	ESE	PR Exam	OR	100	
CAUA31201	ARTIFICIAL INTELLIGENCE	TH	3	2	-	-	-	20	-	-	20	40	20	-	100	4
CAUA31202	DEEP LEARNING	TH	3	2	-	-	-	20	-	-	20	40	20	-	100	4
CAUA31203	BIG DATA ANALYTICS	TH	3	2	-	-	-	20	-	-	20	40	-	20	100	4
CAUA31204	DATA COMMUNICATION AND NETWORKING	TH	2	2	-	-	20	-	20	-	20	40	-	-	100	3
CAUA31205	PROFESSIONAL ELECTIVE I	TH	3	2	-	-	-	-	-	-	-	-	-	-	100	4
CAUA31206	RESEARCH METHODOLOGY AND IPR	CE	2	-	-	-	-	50	-	-	-	-	-	-	50	2
CAUA31207	PROJECT - I	CE	-	4	-	-	30	-	-	-	-	-	-	20	50	2
	Total		16	14	-	-	50	110	20	-	80	160	40	40	600	23

Professional Elective I	
CAUA31205A	FULL STACK SOFTWARE DEVELOPMENT
CAUA31205B	CLOUD DEVELOPMENT OPERATIONS

BOS Chairman

Dean Academics

Director



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T.Y. B.Tech Computer Science and Engineering (Artificial Intelligence)

SEMESTER VI (PATTERN 2020R1)

CourseCode	Course Name	Teaching Scheme (Hrs/Week)				Assessment Scheme (100-mark scale)										Credits
		Course Type	Theory	Lab	Tut	ISA					ESA				Total	
						HA	SCE	PPT	GD	CIE	ESE	PR Exam	OR	TW	100	
CAUA32201	NATURAL LANGUAGE PROCESSING	TH	3	2	-	-	20	-	-	20	40	20	-	-	100	4
CAUA32202	GENERATIVE AI	TH	3	2	-	-	20	-	-	20	40	-	20	-	100	4
CAUA32203	THEORY OF COMPUTATION	TH	3	-	-	20	-	-	-	20	40	-	-	20	100	3
CAUA32204	PROFESSIONAL ELECTIVE II	TH	2	2	-	-	20	-	-	20	40	20	-	-	100	3
	OPEN ELECTIVE I	TH	3	-	1		20	-	-	20	40	-	20	-	100	4
AEC32206	LOGICAL REASONING AND QUANTITATIVE APTITUDE	CE	1	-	-	-	-	-	-	-	-	-	-	50	50	1
CAUA32205	PROJECT II	CE	-	4	-	-	-	-	-	-	-	-	-	50	50	2
M3	MANDATORY COURSE	AU	-	-	-		-	-	-	-	-	-	-	-	-	-
	Total		15	10	1	20	80	-	-	100	200	40	40	120	600	21

Professional Elective II	
CAUA32204A	Soft Computing & Optimization Techniques
CAUA32204B	Cyber Security
AIUA32204D	AIML with mainframe technology

Open Elective I	
IOEUA32205C	Explainable Artificial Intelligence (XAI) for Engineering Applications
IOEUA32205A	Engineering Economics and Fintech
AIUA32204E	Machine Learning Operations (MLOps)

BOS Chairman

Dean Academics

Director



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SEMESTER- V

CAUA31201: ARTIFICIAL INTELLIGENCE

Teaching Scheme	Examination Scheme				
Credits: 04	CIE	SCE	ESE	PR	Total
Lecture (L): 03 hrs / week					
Practical (P): 02 hr / week	20	20	40	20	100
Prerequisites: Linear Algebra, Probability and Statistics, Discrete Mathematics					
Course Objectives: - To understand the various characteristics of intelligent agents. - To learn the different search strategies in AI. - To learn knowledge representation and working knowledge of reasoning in presence of incomplete and uncertain information. - To learn how to represent knowledge in solving AI problems and to introduce the concepts of Expert Systems and Machine Learning. - To know about the various applications of AI. - To know advancements of AI					
Course Outcomes: After completion of the course, student will be able to: - Understand different types of AI Agents and environment. - Implement various AI search algorithms. - Understand fundamentals of knowledge representation and working knowledge of reasoning in presence of incomplete and uncertain information. - Apply knowledge representation, reasoning and machine learning techniques to real world problems and design the Expert Systems. - Apply AI techniques for real world application. - To understand best practices for scalability, reliability, and security in cloud-based AI and ML solutions.					
Contents					
Unit I: Introduction to Artificial Intelligence & Intelligent Agents (6 Hrs)					
Introduction, Definition, Future of Artificial Intelligence, Characteristics of Intelligent Agents, Typical Intelligent Agents, Environment, Environment Types, Problem Solving Approach to Typical AI problems. Defining the problem as a State Space Search, Problem Characteristics, Problem solving Methods, Search Strategies: Uninformed - Informed - Heuristics Constraint Satisfaction Problems, Constraint Propagation, Backtracking and look ahead strategy Search.					
Unit II: Search Strategies (6 Hrs)					
Uninformed Search Strategies: Breadth-first search, Depth-first search Informed (Heuristic) Search Strategies: Greedy best first search, Optimal search: A* search, Minimizing the total estimated solution cost					



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Heuristic search techniques: Heuristic Function, Generate and test, Hill Climbing, Best-First Search, Beam search, Tabu Search.

Unit III: Knowledge Representation and Inference (6 Hrs)

Knowledge representation, Types of knowledge, Structured representation of knowledge, Propositional logic, Predicate logic, First order logic. Rule based expert system, Rules Inference: Backward chaining, Forward chaining, Rule value approach, Inference engine, Rules. Knowledge Representation: Conceptual Dependency, Frames, Semantic nets.

Unit IV: Planning & Learning in Expert Systems (6 Hrs)

Planning: Block world problem, Goal Tree, Non-linear planning, Hierarchical planning, Goal stack planning
Definition of learning, Forms of learning, learning by taking advice, Learning in problem solving, Induction learning, Explanation based learning, Formal learning theory, Ensemble learning.
Expert systems - Intelligent System Vs Expert system, Architecture of expert systems, Roles of expert systems, Knowledge Acquisition, Meta Expert systems shells.

Unit V: AI Applications (6 Hrs)

AI in Speech Recognizer, Natural Language Processing, Chatbots, Computer Vision, Role of AI in Industry 4.0 and industry 5.0.
AI in Robotics – Robotic sensors, Agriculture robots, Domestic robots, Delivery robots, Mining robots, Space robots, Robotics in biological sensing, Unmanned Aerial Vehicle – UAV, Forestry.
AI in E-commerce, Education, Manufacturing, Navigation, Healthcare, Gaming, Social media, Security, Lifestyle

Unit VI: Advancements of AI on Cloud Platforms (6 Hrs)

Amazon Web Services (AWS): Amazon SageMaker, Alexa, Lex, and Polly, Conversational gents, Amazon Comprehend, natural language processing, Amazon Rekognition, image and video, Amazon Translate, Amazon Machine Learning, Amazon Transcribe, transcription, Amazon Textract, document analysis.
Microsoft Azure: Microsoft Azure Machine Learning Studio, Azure Machine Learning Service, Azure Cognitive Services Google Cloud Platform (GCP): AI Hub Google Cloud AI Building Blocks

Textbooks:

1. Stuart Russell, Peter Norvig, Artificial Intelligence: A Modern Approach, Prentice Hall, Fourth edition, 2020
2. Artificial Intelligence by Elaine Rich, Kevin Knight and Nair ISBN-978-0- 07008770-5, TMH
3. Artificial Intelligence by Patrick Henry Winston, Addison-Wesley Publishing Company, ISBN 0-201-53377-4

Reference Books:



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1. Artificial Intelligence with Python -Second Edition-Alberto ArtasanchezPrateek Joshi-Packt Publishing Ltd
2. Artificial Intelligence by Saroj Kausik ISBN:- 978-81-315-1099-5, Cengage Learning
3. Artificial Intelligence and Intelligent Systems by Padhy, Oxford University Press
4. Deepak Khemani, "A First Course in Artificial Intelligence", McGraw Hill Education(India), 2013, ISBN : 978-1-25-902998-1

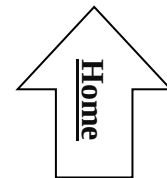
List of NPTEL Courses:

1. https://cse.iitkgp.ac.in/~pallab/artificial_intelligence_autumn_2020/index.html
2. https://onlinecourses.nptel.ac.in/noc22_cs56/preview
3. https://onlinecourses.nptel.ac.in/noc23_ge40/preview

List of Assignments:

Similar problem statements can be floated for each batch by subject in-charge. Implement any 7 of them.

1. Implement DFS, BFS for 8-puzzle problem .
2. Implement constraint satisfaction problem.
3. Perform parsing of family tree using knowledge-base.
4. To Implement A* Algorithm for an application.
5. Implement minmax algorithm for game playing.
6. Implement basic search strategies – 8-Queens Problem
7. Implement Forward Chaining Algorithm
8. Implement backward chaining algorithm
9. Create a Chabot application for any real-world scenario



CAUA31202: DEEP LEARNING

Teaching Scheme	Examination Scheme				
Credits: 04 Lecture (L): 03 hrs / week Practical (P): 02 hrs / week	CIE	SCE	ESE	PR	Total
	20	20	40	20	100
Prerequisites: Probability and Statistics, Machine Learning					
Course Objective(s): 1. To explain the theoretical foundations, algorithms, and methodologies of deep learning. 2. To understand the basics of deep neural networks. 3. Comparing different deep learning architectures. 4. To characterize different architectures and parameters of deep neural networks such as CNN. 5. To explain deep generative models using autoencoders and representation Learning. 6. To describe Reinforcement Learning.					
Course Outcomes: After completion of the course, student will be able to - 1. Analyze the different issues and needs of parameters required for the design of any deep neural network. 2. Apply convolutional neural networks-based techniques and analyze their performance. 3. Apply recurrent neural networks-based algorithms and analyze their performance. 4. To implement and apply deep generative models. 5. To apply deep generative models using autoencoders and representation learning. 6. To understand reinforcement learning process and deep learning applications.					
Contents					
Unit I: Introduction of Deep Learning (8 Hrs)					
Fundamentals of Deep Learning: Introduction to Deep Learning, Neuron, Neural Networks, Types of Deep Learning Networks, Working of Neural Network					
Perceptrons: Single Layer Perceptron, Multilayer Perceptrons (MLPs), Representation Power of MLPs, Working of Perceptron with Weights and Bias, Sigmoid Neurons, Advantage and challenges of deep learning.					
Activation Functions: Activation Functions for Neural Networks, RELU, LRELU, ERELU, Sigmoid, Tanh, Softmax					
Training Neural Networks: Backpropagation and Forward propagation.					
Gradient Descent - Gradient Descent (GD), Vanishing Gradient Problem, Momentum Based GD, Nesterov Accelerated GD, Stochastic GD, AdaGrad, RMSProp					
Hyper parameters: Layer size, Magnitude (momentum, learning rate)					



Unit II: Convolutional Neural Networks (5 Hrs)
Building blocks: Convolution blocks, Convolution Operation, Parameter Sharing, Padding, Strides, ReLU layer, Max Pooling, Fully Connected Layers, Activation functions, Basic Architecture and mathematical modelling of CNN. YOLO for object detection. Pretrained Models, Transfer learning techniques and popular CNN architectures: AlexNet, VGGNet, ResNet, InceptionV3, DenseNet, EfficientNet, DarkNet, TFLite, ImageNet Case study: Image classification using CNN.
Unit III: Recurrent Neural Networks (6 Hrs)
RNN: Architectural Overview, Types of Recurrent Neural Networks, Feed-Forward Neural Networks vs Recurrent Neural Networks Bidirectional RNNs: Encoder-decoder sequence to sequence architectures, Back-propagation through time for training RNN, Long Short-Term Memory (LSTM) Recurrent Neural Networks, Gated Recurrent Unit Networks
Unit IV: Autoencoders and Transformers (6 Hrs)
Autoencoders and Decoders: Types of Autoencoders: Undercomplete Autoencoders, Regularized Autoencoders, Sparse Autoencoders, Stochastic Encoders and Decoders, Denoising Autoencoders, Contractive Autoencoders, Applications of Autoencoders Transformers: Introduction, Architecture, Self-Attention in Transformers, Multi-head attention in Transformers, Applications
Unit V: Deep Generative Models (5 Hrs)
Deep generative model: Introduction, Generative adversarial network (GAN), discriminator network, generator network, types of GAN, Deep Convolutional Adversarial Network, Applications of GAN networks Case Study: GAN for detection of real or fake images
Unit VI: Reinforcement Learning and Deep Learning Applications (6 Hrs)
Reinforcement Learning: Introduction of reinforcement learning, Elements of Reinforcement Learning, Agent Environment Interaction Cycle, Markov Decision Process, basic framework of reinforcement learning, challenges of reinforcement learning, Simple reinforcement learning for Tic-Tac-Toe. Deep Learning Applications: Image Classification, Social Network analysis, Speech Recognition, Recommender system, Natural Language Processing, Chatbots
Text Books:



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1. Ian Goodfellow, Yoshua Bengio, Aaron Courville, "Deep Learning", MIT Press, ISBN: 978-0262035613, 2017.
2. Josh Patterson, Adam Gibson "Deep Learning: A Practitioner's Approach", O'Reilly Media, ISBN-13:978-1491914250, 2017.
3. Charu C. Aggarwal, "Neural Networks and Deep Learning: A Textbook", Springer, ISBN: 978-3-319-94463-0, 2019.
4. Nikhil Buduma, "Fundamentals of Deep Learning Designing Next-Generation Machine Intelligence Algorithms" O'Reilly

Reference Books:

1. Aston Zhang, Zachary C. Lipton, Mu Li, and Alexander J. Smola, "Dive into Deep Learning", 2021.
2. Giancarlo Zaccane, Md. Rezaul Karim, Ahmed Menshawy "Deep Learning with TensorFlow: Explore neural networks with Python", Packt Publisher, 2017.
3. Umberto Michelucci "Applied Deep Learning. A Case-based Approach to Understanding. Deep Neural Networks" Apress, 2018.
4. Antonio Gulli, Sujit Pal "Deep Learning with Keras", Packt Publishers, 2017.
5. Francois Chollet "Deep Learning with Python", Manning Publications, 2017.

List of MOOC / NPTEL and e-Learning Courses:

1. <https://www.my-mooc.com/en/categorie/deep-learning>
2. https://onlinecourses.nptel.ac.in/noc19_cs58
3. https://onlinecourses.nptel.ac.in/noc21_ee23
4. https://onlinecourses.nptel.ac.in/noc23_cs24/
5. https://onlinecourses.nptel.ac.in/noc23_cs56
6. <https://www.analyticsvidhya.com/blog/2021/07/in-depth-explanation-of-recurrent-neural-network>
7. <https://www.automl.org/automl/>
8. Deep Learning for Visual Computing https://onlinecourses.nptel.ac.in/noc22_ee54
9. Deep Learning - IIT Kharagpur https://onlinecourses.nptel.ac.in/noc22_cs22
10. Deep Learning - IIT Ropar https://onlinecourses.nptel.ac.in/noc22_cs35/

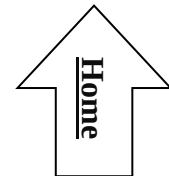
List of Assignments:

Similar problem statements can be floated for each batch by subject in-charge.

1. Implementing Feedforward neural networks in Python using Keras and TensorFlow
2. Facial recognition using OpenCV and deep learning for binary classification.
3. Implement Image classification using convolutional neural networks (CNNs) for multiclass classification.
4. Time series prediction using RNN – stock market analysis or weather forecasting
5. Text identification using OpenCV, Tesseract (OCR) and deep neural network
6. Sentiment analysis using LSTM network or GRU.
7. Object detection using YOLO and Pretrained Model

Mini project is to be performed in a group of 2 to 4 students.

Develop a mini project in a group using different predictive models techniques to solve any real life problem.



CAUA31203: BIG DATA ANALYTICS

Teaching Scheme	Examination Scheme				
Credits: 04	CIE	SCE	ESE	PR	Total
Lecture (L): 03 hrs / week	20	20	40	20	100
Practical (P): 02 hr / week					
Prerequisites: Programming and Problem solving, Data Structures & Algorithms, Operating Systems, DBMS, Machine Learning					
1. To comprehend Big data and its features. 2. To develop various applications based on distributed computing frameworks. 3. To apply various operations on distributed storage. 4. To learn and analyze how to handle large data sets in main memory. 5. To learn spark streaming and programming. 6. To learn the various machine learning techniques for Big Data Analytics.					
Course Outcomes: After completion of the course, student will be able to - 1. Comprehend Big data and its features. 2. Develop various applications based on distributed computing frameworks. 3. Apply various operations on distributed storage. 4. Explore and analyze large data sets in main memory. 5. Make Use of Spark framework and perform streaming and programming. 6. Demonstrate various machine learning techniques for Big Data Analytics.					
Contents					
Unit I: Understanding Big Data (6 Hrs)					
Overview: Classification of Digital Data, Structured and Unstructured Data, Big Data & Big data Computing, Role of Distributed System in Big Data Characteristics of Big Data: 3 Vs and more, Big data Evolution – Definition - Challenges with Big Data. Big Data Applications & Use cases: Web search, Marketing and advertising, Information retrieval, Patterns for Big Data Deployment, IT for IT Log Analytics, The Fraud Detection Pattern					
Unit II: Hadoop Map-Reduce Framework (6 Hrs)					
Basics of functional programming, Map reduce fundamentals, Data flow (Architecture), Architecture of Hadoop, Requirements of Hadoop Framework, Design principle of Hadoop, Comparison with other system, Hadoop Components – Hadoop 1 vs Hadoop 2 – Hadoop Daemon's, Map Reduce Programming: I/O formats, Map side join, Reduce Side Join, Secondary sorting, Pipelining MapReduce jobs, Hadoop Ecosystem overview					
Unit III: Hadoop Distributed File System (6 Hrs)					
Distributed System Overview, Distributed File System with NFS, Basic HDFS Architecture: NameNode, Secondary NameNode, DataNode, checkpoint Node. Features of HDFS, HDFS Assumption and Goals: Hardware failure, Streaming data access, Large datasets, coherency model. HDFS Read and Write Operations.					



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Unit IV: Spark Programming (6 Hrs)

Spark Basics: Introduction to Spark, Spark components, Spark Architecture: Spark client, driver, Spark context, Master and Executors, Spark execution: Job, Stage and Task, Spark Application developments, Spark Streaming: Batch processing and Real time processing, Concepts of Spark streaming, Stream sources, Architecture of Spark streaming, Stateless transformation, Stateful transformation. Spark Programming in Python/Java - Application Execution.

Unit V: High Performance Data Analytics with CUDA (6 Hrs)

Introduction to GPU Computing, CUDA Programming Model, CUDA API, Simple Matrix Multiplication in CUDA, CUDA Memory Model, Shared Memory Matrix Multiplication, Additional CUDA API Features. Data Analysis with Spark Shell.

Unit VI: Machine Learning Algorithms for Big Data Analytics (6 Hrs)

Introduction, Estimating the relationships, Outliers, Variances, Probability Distributions, and Correlations, Regression analysis, Finding Similar Items, Similarity of Sets and Collaborative Filtering, Publicly Available Datasets: UCR STAR [<https://star.cs.ucr.edu>], Twitter Streaming API, Big-data Libraries: MLlib (Machine Learning), GraphX

Text Books:

1. Raj Kamal and Preeti Saxena, "Big Data Analytics Introduction to Hadoop, Spark, and Machine-Learning", McGraw Hill Education, 2018 ISBN: 9789353164966, 9353164966
2. Douglas Eadline, "Hadoop 2 Quick-Start Guide: Learn the Essentials of Big Data Computing in the Apache Hadoop 2 Ecosystem", 1 stEdition, Pearson Education, 2016. ISBN13: 978-9332570351
3. Data Analytics: The Ultimate Beginner's Guide to Data Analytics Paperback – 12 November 2017by Edward Mize.

Reference Books:

1. Hadoop: The Definitive Guide, Tom White, THIRD EDITION, OREILLY
2. Hadoop in Action, Chuck Lam, Manning
3. Professional Hadoop, Benoy Antony et al, July 2016, Wiley
4. Big Data, Big Analytics: Emerging Business Intelligence and Analytic Trends for Today's Businesses", Michael Minelli et al Wiley, 2013.
5. Professional CUDA® C Programming, John Cheng, John Wiley & Sons, Inc.

List of Assignments:

1. Installation and Configuration of Single Node Hadoop System.
2. Configuration of Multi-node Hadoop System.
3. HDFS commands such as Adding files and directories, retrieving files and Deleting files
4. Develop a MapReduce program to calculate the frequency of a given word in a given file. (word count)
5. Develop a MapReduce program to implement Matrix Multiplication.
6. Develop a MapReduce to analyze weather data set and print whether the day is shinny or cool day.



7. Develop a MapReduce program to find the frequency of books published each year and find in which year maximum number of books were published using the following data.
Title Author Published-year Author-country Language No of pages
8. Develop a MapReduce program to analyze Uber data set to find the days on which each basement has more trips using the following dataset.
The Uber dataset consists of four columns they are:
dispatching_base_number date active_vehicles trips
9. Implement applications using SPARK.

CAUA31204: DATA COMMUNICATION AND NETWORKING

Teaching Scheme	Examination Scheme					
Credits:3 Lecture (L): 02 Practical (P): 02 hrs/Week	CIE	PPT	SCE	ESE	OR	Total
	20	20	-	40	20	100
Prerequisites: Digital Electronics and Computer Architecture Course objectives: 1. To understand the fundamental concepts of Data Communication. 2. To classify the transmission media and switching Techniques. 3. To Design and Develop the LAN Topologies. 4. To illustrate the functions of OSI and TCP/IP Protocol. 5. To understand the functionality of Data link layer						
Course Outcomes: Upon completion of the course, students will be able to 1. Elaborate the concepts of Data Communication. 2. Discriminate various Transmission Media, Connecting devices and Switching Techniques for digital data communication. 3. Demonstrate the functions of OSI layers & TCP/IP protocol stack. 4. Comprehensive functions of Error Control and Data Link Control						
Contents						
Unit I – Introduction to Data Communications (6 Hrs)						
Process of Data Communication and its components: Transmitter, Receiver, Medium, Message, Protocol. Analog and Digital: Analog and Digital Data, Analog and Digital Signals, Periodic and Non-periodic Signal. Transmission Impairment: Attenuation, Distortion, Noise Data Rate Limits: Noiseless Channel: Nyquist Bit Rate, Noisy Channel: Shannon Capacity, Using Both Limits Performance: Bandwidth, Throughput, Latency (delay), Bandwidth-delay Product, Jitter. Modes of Communication (Simplex, Half Duplex, Full Duplex).						
Unit II – Transmission Media and Switching (6 Hrs)						
Guided Media: Twisted-Pair, Coaxial and Fiber-Optic Cable Unguided Media: Radio Waves, Microwaves, Infrared Connecting devices: Hubs, Repeaters, Active hubs, Bridges, Switches, Routers, Gateways Switching: Circuit-switched Networks: Three Phases, Efficiency, Delay. Packet switching: Datagram networks, Virtual circuit networks.						
Unit III – Layered Model and Topologies (6 Hrs)						



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The OSI Model: Layered Architecture, peer-to-peer Processes, Encapsulation of Layers in the OSI Model, TCP/IP Protocol Suite, LAN, MAN, WAN, Topologies- star, mesh, bus, hybrid etc.
Addressing: Physical & logical Addresses, Port Addresses, Specific Addresses

Unit IV– Error Control and Data Link Control (6 Hrs)

Types of errors: Single bit error and burst error, Redundancy, detection versus correction, forward error correction versus retransmission,
Block coding: error detection, error correction, CRC, polynomial, checksum, hamming code, hamming Distance, DLC Services: Framing, Flow and error control DLL Protocols: Simple protocol, Stop n wait, Go back to N, Selective repeat Point-to-point Protocol (PPP) : Framing, Transition Phases, Multiplexing, Multilink PPP.

Textbooks:

1. Fourauzan B., "Data Communications and Networking", 5th edition, McGraw-Hill Publications
2. Stallings William., "Data and Computer Communications", Sixth Edition, Prentice Hall of India.

Reference books:

1. Andrew S. Tenenbaum , "Computer Networks", Pearson
2. Douglas E. Comer, "Computer Networks and Internets", Pearson Education
3. Godbole Achyut, "Data Communication and Networks", Tata McGraw Hill Publication.

List of Assignments:

1. Exploring different Analog and Digital Data conversion techniques.
2. Network Representation on Network Simulator tool
3. Study of DCE, DTE, straight and cross cables
4. Identify MAC and IP addresses (physically as well as using packet tracer tool).
5. Demonstrate the use of Wireshark tool for network analysis
6. Setup a wired LAN using Layer 2 Switch. It includes preparation of cable, testing of cable using line tester, configuration machine using IP addresses, testing using PING utility and demonstrating the PING packets captured traces using Wireshark Packet Analyzer Tool.
7. Demonstrate the different types of topologies and types of transmission media by using a packet tracer tool.
8. Test with basic network tools. Read the man pages of ifconfig, ping, traceroute, arp, dig and nslookup. Write the solution of the following problems into the report and also paste the screenshots of Terminal into the report.
 - (a) Find the IP address of your machine using 'ifconfig' command.
 - (b) Use 'ping' command and find out (i.) the average RTT(round trip time). (ii.) the %packet loss. (iii.) size of packet that is sent to www.iitk.ac.in server. (iv.) size of packet that is received by your machine.
 - (c) Use 'dig www.iitk.ac.in' command and find out i. the ip address of iitk.ac.in. ii. the ip addresses of local DNS servers of your institute



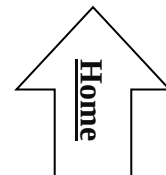
(d) Use 'tracert www.iitk.ac.in' and find out i. number of hops in between your machine and iitk.ac.in server. ii. the ip address of your network gateway of your subnet.

(e) Use 'arp -an' command to find out the MAC address of the device that is performing as your network gateway.

9. Write a program for hamming code error control method.
10. Write a program for checksum stop-n-wait flow control method.
11. Write a program for checksum Go-back-N flow control method.

NPTEL / MOOC course:

1. "Data Communication", IIT Kharagpur Prof. Ajit Pal
<https://nptel.ac.in/courses/106105082>



CAUA31205: PROFESSIONAL ELECTIVE I

Teaching Scheme		Examination Scheme				
Credits:3		CIE	SCE	ESE	OR	Total
Lecture (L): 02						
Practical (P): 02 hrs/Week		20	20	40	20	100
Course Name	Application Developer	IBM Full Stack Software Developer				
Full Stack Software Development (CAUA31205A)	IBM Full Stack Software Developer	Introduction to Cloud Computing				
	IBM Full Stack Software Developer	Introduction to Web Development with HTML, CSS, JavaScript				
	IBM Full Stack Software Developer	Getting Started with Git and GitHub				
	IBM Full Stack Software Developer	Developing Cloud Native Applications				
	IBM Full Stack Software Developer	Developing Cloud Apps with Node.js and React				
	IBM Full Stack Software Developer	Introduction to Containers w/ Docker, Kubernetes & OpenShift				
	IBM Full Stack Software Developer	Python for Data Science, AI & Development				
	IBM Full Stack Software Developer	Application Development using Microservices and Serverless				
	IBM Full Stack Software Developer	Full Stack Cloud Development Capstone Project				
	IBM Full Stack Software Developer	Full Stack Software Developer Assessment				
2	DevOps Engineer	IBM DevOps and Software Engineering				
Cloud Development Operations (CAUA31205B)	IBM DevOps and Software Engineering	Introduction to DevOps				
	IBM DevOps and Software Engineering	Introduction to Cloud Computing				
	IBM DevOps and Software Engineering	Introduction to Agile Development and Scrum				
	IBM DevOps and Software Engineering	Hands-on Introduction to Linux Commands and Shell Scripting				
	IBM DevOps and Software Engineering	Getting Started with Git and GitHub				
	IBM DevOps and Software Engineering	Python for Data Science, AI & Development				
	IBM DevOps and Software Engineering	Introduction to Containers w/ Docker, Kubernetes & OpenShift				
	IBM DevOps and Software Engineering	Application Development using Microservices and Serverless				



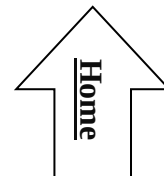
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	IBM DevOps and Software Engineering	Introduction to Test Driven Development (TDD)
	IBM DevOps and Software Engineering	Continuous Integration and Continuous Delivery (CI/CD)
	IBM DevOps and Software Engineering	Application Security and Monitoring
	IBM DevOps and Software Engineering	DevOps Capstone Project

**CAUA31206: RESEARCH METHODOLOGY AND IPR**

Teaching Scheme	Examination Scheme					
Credits: 02 Lecture (L): 02 hrs / week	CIE	ISE	SCE	ESE	TW	Total
	--	--	50	--	--	50
Prerequisites: Nil						
Course Objective(s): 1. Explain the formulation of Research Problem 2. Explain the importance of ideas, concept and creativity 3. Transfer the knowledge about the IPR required for Engineer's 4. Describe how IPR creates National wealth 5. Teach National and International IP System						
Course Outcomes: Upon completion of course, students will be able to: 1. Formulate the research problem with appropriate objectives 2. Understand the right of ownership, scope of protection as well as the ways to create and to extract value from IP. 3. Identify different types of Intellectual Properties (IPs) 4. Discover how IPR are regarded as a source of national wealth and mark of an economic leadership in context of global market scenario 5. Analyze national & International IP system.						
Contents						
Unit I: Introduction to Research Problem (6 Hrs)						
Meaning of research problem, Sources of research problem, Criteria Characteristics of a good research problem, Errors in selecting a research problem, Scope and objectives of research problem. Approaches of investigation of solutions for research problem, data collection, analysis, interpretation, Necessary instrumentations, Effective literature studies approaches, analysis Plagiarism, Research ethics.						
Unit II: Introduction to Intellectual Property (6 Hrs)						
Introduction to the concepts Property and Intellectual Property, Nature and Importance of Intellectual Property Rights Understanding the types of Intellectual Property Rights: - Patents, Designs, Trademarks (Registered and unregistered trademarks), Copyright, Traditional Knowledge, Geographical Indications, Trade Secrets, Idea Patenting, (Case Studies)						
Unit III: Patent Formulation (6 Hrs)						
New Developments in IPR, Process of Patenting and Development: technological research, innovation, patenting, development, International Scenario: WIPO, TRIPs, Indian Patent Office and its Administration.						



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Unit IV: Patent Act and Licensing (6 Hrs)

Administration of Patent System–Patenting under Indian Patent Act, Patenting under PCT, Patent Rights and its Scope, Licensing and transfer of technology, Patent information and database. Provisional and Non-Provisional Patent Application and Specification

Text Books:

1. Resisting Intellectual Property by Halbert, Taylor&Francis Ltd, 2007
2. Industrial Design by Mayall, Mc Graw Hill
3. Intellectual Property in New Technological Age by Robert P. Merges, Peter S. Menell, Mark A. Lemley

Reference Books:

1. Wayne Goddard and Stuart Melville, “Research Methodology: An Introduction”
2. Ranjit Kumar, 2nd Edition, “Research Methodology: A Step by Step Guide for beginners”
3. Intellectual Property Rights under WTO by T. Ramappa, S. Chand
4. Introduction to Design by Asimov, Prentice Hall

CAUA31207: PROJECT - I

Teaching Scheme	Examination Scheme					
Credits: 02 Practical (P): 04 hrs / week	CIE	ISE	SCE	OR	TW	Total
	--	--	--	20	30	50
Prerequisites:						
Course Objective(s): - To apply the learnt skill sets required to solve an engineering related problem. - To expose students to product development cycle using industrial experience, use of state of art technologies. - To provide the students with the opportunity to design or conduct independent research related to IT courses. - To encourage and expose students for participation in research paper presentation activities and IPR. - To learn the importance of teamwork, documentation and ethics.						
Course Outcomes: Upon completion of course, students will be able to: - Identify the problem that needs technological solution and define scope. - Demonstrate a sound literature review and design the solution. - Implement the solution and test by applying knowledge and various tools. - Track the progress of the project with the help of a project management tool. - Perform effectively as an individual and a team member in multi-disciplinary teams. - Communicate with engineers and the community at large in written and oral forms.						
Contents						
Unit I: - Title and scope finalization (6 Hrs)						
Problem statement: a concise statement of the problems which initiate the industrial problem or research questions or design ideas. Answering questions: What is the issue that we want to address (problem or question)? Why need to address the issues? How can the project solve the issues?, d. Who will benefit from the project? Setting a clear goal of what needs to be accomplished by doing the work. Students should						
Objectives: only state the technical objective of the project (e.g. to evaluate the performance of the design, to test a hypothesis, to study the relationship between variable x and variable y etc.). Use measurable action verbs when defining an objective (e.g. define, design, identify, describe, analyze, evaluate etc).						
Scope: Scope sets a clear boundary (time, geography, environment, function etc.) of our work to provide a common understanding of the project among stakeholders (in FYP the stakeholders are students, lecturer, panels etc.). Scope makes our project achievable and realistic by defining the limits and constraints of the study.						
Formal technical review-0 (FTR-0)						



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Unit II: Literature review and design (6 Hrs)

Literature review: To summarize and synthesize the ideas of others (Similar systems and research work) Discuss the techniques or different design decisions; make comparisons and give technical comments; Summarize the comparison with tables or graphs to clarify the differences.

Design: SRS; Use of the designing tools; discussion on the tradeoff of a particular design decision; Selection of Technology and Tools, UI/UX design, database design, design of test cases and performance metrics Installations; Use of various tools

Formal technical review-1 (FTR-1)

Unit III: Implementation and Testing (6 Hrs)

Implementation of algorithms, databases and modules as per design; Testing: Analysis of results and performance; comparative analysis (comparison with existing/ known algorithms/ systems) and validation of results; discussions and conclusions.

Formal technical review-2 (FTR-2)

Unit IV: Report writing and Project outcome generation (6 Hrs)

Documentation and presentation: Project report writing and demo video creation. The student shall prepare and submit the report of Project work in standard format for satisfactory completion of the work that is duly certified by the concerned guide and head of the Department/Institute.

Project outcome: Submit the research paper based on the work done to a reputed conference/ journal or IPR or participation in hackathon or participation in project competition as per the suggestions of the guide. For industry sponsored projects, the work must be submitted/ deployed as per industry requirement and obtain the completion certificate from the industry.

Project Oral exam: The assessment will be broadly based on work undergone, content delivery, presentation skills, documentation, question-answers and report.



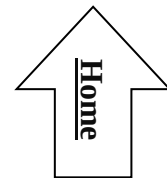
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SEMESTER- VI



CAUA32201: NATURAL LANGUAGE PROCESSING

Teaching Scheme	Examination Scheme				
Credits: 04 Lecture (L): 03 hrs / week Practical (P): 02 hr / week	CIE	SCE	ESE	PR	Total
	20	20	40	20	100
Prerequisite: Theory of Computation, Compiler Design and Basic Understanding of Probability Theory					
Course Objectives: - 1. To be familiar with fundamental concepts and techniques of natural language processing (NLP). 2. To acquire the knowledge of various morphological, syntactic, and semantic NLP tasks 3. To develop the various language modelling techniques for NLP 4. To use appropriate tools and techniques for processing natural languages 5. To Describe Applications of NLP and Machine Translations 6. To comprehend the advance real-world applications in NLP domain					
Course Outcomes: 1. Describe the fundamental concepts of NLP, challenges and issues in NLP 2. Analyse Natural languages morphologically, syntactical and semantically 3. Illustrate various language modelling techniques 4. Integrate the NLP techniques for the information retrieval task 5. Apply the scientific process for machine transliteration, machine translation and information retrieval for given natural language using statistical approach 6. Demonstrate the use of NLP tools and techniques for text-based processing of natural languages.					
Unit I: Introduction to Natural Language Processing			(6 Hrs)		
Contents: Introduction: Scope, Applications and Challenges. Brief history and evolution of NLP: Programming languages Vs Natural Languages, Are Natural Languages Regular? Finite automata for NLP, Stages of NLP, NLP Basics of text processing: Tokenization, Stemming, Lemmatization, Part of Speech Tagging; NLP Components: Syntax, Semantics, and Pragmatics; Introduction to Regular Expressions and their Applications in NLP.					
Unit II: Language Syntax and Semantics			(6 Hrs)		
Contents: Morphological Analysis: What is Morphology? Types of Morphemes, Inflectional morphology & Derivational morphology, Morphological parsing with Finite State Transducers (FST) Syntactic Analysis: Syntactic Representations of Natural Language, Parsing Algorithms,					



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Probabilistic context-free grammars, and Statistical parsing Semantic Analysis: Lexical Semantic, Relations among lexemes & their senses Homonymy, Polysemy, Synonymy, Hyponymy, WordNet, Word Sense Disambiguation (WSD), Dictionary based approach, Latent Semantic Analysis	
Unit III: Language Modelling	(6 Hrs)
Contents: Probabilistic language modelling, Markov models, Generative models of language, Log-Liner Models, Graph-based Models N-gram models: Simple n-gram models, Estimation parameters and smoothing, Evaluating language models, Word Embeddings/ Vector Semantics: Bag-of-words, TFIDF, word2vec, doc2vec, Contextualized representations (BERT) Topic Modelling: Latent Dirichlet Allocation (LDA), Latent Semantic Analysis, Non Negative Matrix Factorization	
Unit IV: Information Retrieval using NLP	(6 Hrs)
Contents: Information Retrieval: Introduction, Vector Space Model Named Entity Recognition: NER System Building Process, Evaluating NER System Entity Extraction, Relation Extraction, Reference Resolution, Coreference resolution, Cross Lingual Information Retrieval	
Unit V: Machine Translation	(6 Hrs)
Contents: Introduction, Problems in MT, Characteristics of Indian Languages, Machine Translation Approaches, Direct Machine Translation, Rule-Based MT, Corpus Based Machine Translation, Semantic/Knowledge based MT Systems, Translation involving Indian Languages, Statistical-Based using MT Tools - GIZA++, SRTLM and Moses, Evaluation Metrics	
Unit VI: NLP Tools and Techniques	(6 Hrs)
Contents: Prominent NLP Libraries: Natural Language Tool Kit (NLTK), spaCy, TextBlob, Gensim etc. Linguistic Resources: Lexical Knowledge Networks, WordNets, Indian Language WordNet (IndoWordnet), VerbNets, PropBank, Treebanks, Universal Dependency Treebanks Word Sense Disambiguation: Lesk Algorithm WordNets for Word Sense Disambiguation.	
Text Books: 1. Jurafsky, David, and James H. Martin Speech and Language Processing: An Introduction to Natural Language Processing , Computational Linguistics and Speech PEARSON Publication 2. Manning, Christopher D., and nrich Schütze Foundations of Statistical Natural Language Processing, Cambridge, MA: MIT Press	
Reference Books: 1. Steven Bird, Ewan Klein, Edward Loper Natural Language Processing with Python Analysing Text with the Natural Langua Publication 2. Dipanjan text Analytics with Python: A Practical Real-World Approach to Gaining Actionable Insights from your Publication ISBN: 9781484223871	



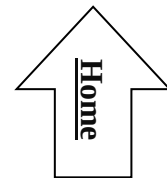
3. Alexander Clark, Chris Fox, and Shalom Lappin The Handbook of Computational Linguistics and Natural Language Processing, Wiley Blackwell Publications
4. Jacob Eisenstein Natural Language Processing, MIT Press
5. Jacob Eisenstein An Introduction to Information Retrieval Cambridge University Press

E-Learning:

1. www.nfnlp.com
2. <https://www.my-mooc.com/en/mooc/natural-language-processing-nlp/>
3. <https://www.mooc-list.com/tags/nlp>
4. <https://huggingface.co/learn/nlp-course/>
5. <https://www.coursera.org/learn/attention-models-in-nlp>

List of Assignments:

1	Perform tokenization (Whitespace, Punctuation-based, Treebank, Tweet, MWE) using NLTK library. Use porter stemmer and snowball stemmer for stemming. Use any technique for lemmatization.
2	Perform bag-of-words approach (count occurrence, normalized count occurrence), TF-IDF on data. Create embeddings using Word2Vec
3	Perform text cleaning, perform lemmatization (any method), remove stop words (any method), label encoding. Create representations using TF-IDF. Save outputs
4	Create a transformer from scratch using the Pytorch library
5	Morphology is the study of the way words are built up from smaller meaning bearing units. Study and understand the concepts of morphology by the use of add delete table.
6	Use a more advanced model for sentiment analysis
7	Write a better auto-complete algorithm using an N-gram model (similar models are used for translation, determining the author of a text, and speech recognition)
8	Mini Project Fine tune a pre-trained transformer for any of the following tasks on any relevant dataset of your choice: ☐ Neural Machine Translation ☐ Classification ☐ Summarization 2. POS Taggers For Indian Languages 3. Feature Extraction using seven moment variants 4. Feature Extraction using Zernike Moments. Communicate project outcomes through a detailed report/article and a professional presentation.



CAUA32202: GENERATIVE AI

Teaching Scheme	Examination Scheme				
Credits: 04	CIE	SCE	ESE	OR	Total
Lecture (L): 03 hrs / week					
Practical (P): 02 hr / week	20	20	40	20	100
Prerequisite: Machine Learning, Deep Learning, Artificial Intelligence, Probability and statistics, Linear Algebra, python programming.					
Course objectives: 1. To understand Generative AI- its evolution and capabilities. 2. To build correct Prompts using prompt engineering tools 3. To understand and implement Generative AI models 4. To Analyze the different capabilities of Generative AI & work on Gen AI Platforms 5. To understand and implement applications of Generative AI and Different Tools 6. To analyze limitations, ethical, social and legal dimensions of Generative AI					
Course Outcomes: Upon completion of the course, students will be able to 1. Interpret Concepts of Generative AI 2. Build powerful prompts 3. Implement Generative AI models and analyze their performances 4. Analyze the capabilities of Generative AI 5. Implement different Generative AI tools and understand real life application. 6. Analyze limitations, ethical, social and legal dimensions of Generative AI					
Unit I: Introduction to Generative AI			(6 Hrs)		
Overview of Generative AI, Generative Pre-trained Transformer motivation, Evolution of Generative Models, Generative models vs. Discriminative models					
Types of Generative AI models - Variational Autoencoders (VAEs), Generative Adversarial Networks (GANs), Multimodal / Diffusion Models					
Applications of GenAI : DALL-E, Midjourney, ChatGPT, Named Entity Recognition, Natural Language Processing Tasks, Emerging trends – Entertainment, Finance, Banking, Manufacturing, Healthcare, Real-Estate, Supply-chain logistics, E-commerce, Education, Hospitality, Fashion.					
Unit II: Prompt Engineering			(6 Hrs)		
Prompt, Structuring a Prompt, Correct and Incorrect Prompts, Building Blocks of Well-Constructed Prompts, Task Formulation in Prompts, Prompt Patterns, Prompt Tuning Techniques, Fine-tuning Prompts for Specific Tasks, Domain-specific Prompt Engineering.					



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Prompt Engineering Approaches - Interview Pattern, Chain of Thought Pattern, Tree of Thought Pattern Techniques for skilfully crafting prompts - Task specification, Contextual guidance, Zero Shot Prompting, Few-shot prompting	
Unit III: Generative AI and LLM Frameworks	(6 Hrs)
Open-source Tools and Resources for Generative AI Tensorflow and PyTorch, Hugging Face, LangChain, LlamaIndex, Generative AI providers - OpenAI, Cohere, Anthropic, LLMFlow LLMs available: GPT, LLaMA, LaMDA, PaLM, Gemini, Falcon, BLOOM, Evaluation Metrics for LLMs	
Unit IV: Capabilities of Generative AI	(6 Hrs)
Models and Tasks: Text Generation, Audio Generation, Video Generation, Code Generation, Data Generation and Augmentation. 3-D Shape Generation. Audio Generation – Creating Music, Text to speech, Speech to Speech Image Generative Models - Generative Adversarial Networks (GANs), Code Generative Tools - Amazon CodeWhisper, OpenAI Codex Retrieval-Augmented Generation (RAG)s	
Unit V: Vector Databases and Search	(6 Hrs)
Vector Databases, High-dimensional Data Storage, Vector Embeddings, High-Dimensional Semantic Similarity, Personalized Search, Multimodal Search, Knowledge Graph Search. Semantic search, Conversational search, Visual search, Personalization and Relevance Ranking, Evaluation of Search Systems	
Unit VI: Security, Ethics and Bias in Generative AI	(6 Hrs)
Deepfakes and Misinformation, Privacy and Data Security, Privacy-Preserving Generative Models, Explainability and Transparency, Bias Detection and Mitigation, Fairness, Bias, Copyright Issues, Model Auditing; Social and Economic impacts of Gen AI; Responsible AI.	
Text Books: 1. Choi, H., Kim, J. (2023). Generative AI: A Practical Guide to Applications and Ethics. Springer. 2. Jozefowicz, R., Zaremba, W., Sutskever, I. (2023). An Introduction to Generative Models with Applications. Cambridge University Press. 3. Liu, L., Xu, H. (2020). Generative deep learning: Teaching machines to paint, write, compose, and play. O'Reilly Media.	
Reference Books: 1. Ramesh, A., Pavlov, M., Goh, G., et al. (2022). DALL·E: Creating images from text. OpenAI. Retrieved from https://openai.com/research/dall-e	



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2. Lee, J., Kim, J. (2023). Generative AI for Software Engineering: Techniques and Tools. CRC Press.

E-Learning:

1. Swayam Course on Generative AI and Large Language Models, By Naveen Kumar Bhansali, Indian Institute of Management Bangalore (IIMB).
2. OpenAI. (n.d.). Research and blog on AI advancements. Retrieved from <https://www.openai.com/blog>.
3. Ramesh, A., Pavlov, M., Goh, G., et al. (2022). DALL·E: Creating images from text. OpenAI. <https://openai.com/research/dall-e>

List of Assignments: (Any 8)

1	Implement a basic probabilistic generative model (e.g., Gaussian Mixture Models). OR Build a simple text generator using Markov Chains.
2	Implement all Prompt Engineering approaches- Interview Approach, COT, TOT. Compare and contrast all and analyse their applications. Implement Zero shot and Few shot prompting and compare their results.
3	Fine-tune GPT or GPT-2 for creative story generation.
4	Build a simple question-answering chatbot using a pre-trained language model.
5	Generate creative artwork using Stable Diffusion. OR Implement and modify a pre-trained diffusion model for a custom dataset.
6	Create a text-to-image generation pipeline using a pre-trained model like DALL-E.
7	Implement a basic multimodal system for image captioning using CLIP.
8	Develop a text-to-speech model using Tacotron or WaveNet.
9	Use an AI-based tool to generate simple video animations.
10	Create images based on text prompts using a text-to-image model to explore Transformer capabilities in multimodal tasks.
	Mini Project Develop a practical Generative AI system addressing a real-world challenge or explores a creative application of generative AI. Guidelines: • Teams of 2–4 students. • Formulate a project proposal based on identified needs or creative ideas in the field of Gen AI. • Collaborate effectively in teams to execute a Gen AI project.



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- | | |
|--|---|
| | <ul style="list-style-type: none">• Implement, Analyze and Evaluate a Gen AI model using appropriate methodologies and tools. |
|--|---|

	Communicate project outcomes through a detailed report/article and a professional presentation.
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CAUA32203: THEORY OF COMPUTATION

Teaching Scheme	Examination Scheme				
Credits: 03 Lecture (L): 03 hrs / week	HA	CIE	TW	ESE	Total
	20	20	20	40	100
Prerequisite: Basic Mathematics, Discrete Structure.					
Course objectives: 1. Understand fundamentals of computer mathematics. 2. Study the various abstract computing models 3. Study the Regular expression 4. Study the different types of languages& their relationships. 5. Study the different types of grammar & ambiguity in the grammar 6. Study the recursive & recursively enumerable languages					
Course Outcomes: After completion of the course, student will be able to 1. Use the mathematical preliminaries with the help of proofs and lemmas for language derivation. 2. Understand Regular expression, its conversion to automata and its applications. 3. Understand different types of languages, grammars and removal of ambiguity in the grammar. 4. Construct pushdown automata for formal language and its applications. 5. Understand the concept of language acceptability by Turing machine and types of Turing machine. 6. Understand the recursive & recursively enumerable languages.					
Unit I: Theory of Automata			(6 Hrs)		
Basic Mathematical Objects: Sets, Logic, functions, Relations, Strings. Definition of Finite Automata (FA), Description of FA, Transition Systems, Acceptability of a String by a FA, Deterministic and Non deterministic FA, Equivalence of DFA and NFA, FA with output: Moore and Mealy machines-Definitions, Models, inter-conversion, minimization of FA. Applications of Finite Automata.					
Unit II: Regular Expressions (RE) and Languages			(6 Hrs)		
Recursive definition of regular expression, regular set, identities of regular expressions, Conversion of NFA With epsilon moves to DFA, Conversion-RE to DFA , Conversion-DFA to RE, Equivalence of R.E, Equivalence of FA, Arden’s theorem, Pumping lemma for regular languages, Closure properties of regular languages, Applications of R.E					
Unit III: Grammar			(6 Hrs)		
Grammar- Definition, representation of grammar, Chomsky hierarchy, Context Free Grammar- Definition, Derivation, sentential form, parse tree, inference, derivation, parse tree, ambiguity in grammar and					



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language, Simplifications of context free Grammar-Eliminating unit productions, useless symbols, and Null-productions, Normal Forms for CFG- Chomsky normal form.	
Regular Grammar- Definition, left linear, right linear grammar, Inter conversion between left linear and right linear regular grammar Application of CFG: Parser, Markup languages	
Unit IV: Push Down Automata	(6 Hrs)
Push Down Automata- Definition, Notation, acceptance by final state, acceptance by empty stack, Equivalence of PDA and CFG- Grammar to PDA, PDA to Grammar, Deterministic PDA and Non DeterministicPDA. Closure Properties of CFLs.	
Unit V: Turing machines	(6 Hrs)
Turing machine Model, Representation of Turing machine, Language acceptability by Turing machine, Design of Turing machine, Types of TM, Halting Problem, Applications of Turing Machine.	
Unit VI: Recursively Enumerable Languages	(6 Hrs)
Recursively Enumerable and Recursive, Properties of recursive and recursively enumerable languages, Post correspondence problem, The Classes P and NP: Problems Solvable in Polynomial Time, An Example: Kruskal's Algorithm, Nondeterministic Polynomial Time, An NP Example: The Traveling Salesman Problem.	
Undecidability: Reducing one problem to another, Other unsolvable Problems. Tractable and In Tractable Problems	
Text Books: 1. Mishra K., Chandrasekaran N., 'Theory of Computer Science (Automata, Languages and Computation)', Second Edition, Prentice Hall of India 2. John C Martin. "Introduction to Language and Theory of Computation", Third edition, Tata McGraw- Hill	
Reference Books: Hopcroft J., Motwani R., Ullman J., "Introduction to Automata Theory, Languages and Computations", Third edition, Pearson Education Asia.	
E-Learning: 1. https://onlinecourses.nptel.ac.in/noc25_cs70/preview 2. https://onlinecourses.nptel.ac.in/noc17_cs34	

CAUA32204A: SOFT COMPUTING AND OPTIMIZATION TECHNIQUES

Teaching Scheme	Examination Scheme				
Credits: 03 Lecture (L): 03 hrs / week	CIE	SCE	ESE	PR	Total
	20	20	40	20	100
Prerequisites: Linear Algebra, Data Structure, Machine Learning					
Course Objective(s): 1. To differentiate between traditional hard computing techniques and soft computing methods in problem-solving. 2. To implement soft computing techniques to solve real-world optimization problems effectively. 3. To explore heuristic and metaheuristic optimization methods like genetic algorithms, simulated annealing, and particle swarm optimization. 4. To understand the basics of optimization problems, including mathematical formulations, constraints, and objective functions. 5. To analyze different optimization algorithms and their applications in various domains.					
Course Outcomes(s): Upon completion of course, students will be able to: 1. Learn soft computing techniques and their applications. 2. Analyze various neural network architectures. 3. Define the fuzzy systems. 4. Understand the genetic algorithm concepts and their applications. 5. Identify and select a suitable Soft Computing technology to solve the problem; construct a solution and implement a Soft Computing solution.					
Contents					
Unit I: Introduction to Soft Computing					(6 Hrs)
Introduction of soft computing, soft computing vs. hard computing, various types of soft computing techniques, Basic toolsof soft computing – Fuzzy logic, neural network, evolutionary computing. Intelligent Agents: Agents and Environments, Structure of Agent, types of Agent, Some applications of Soft computing techniques.					
Unit II - Fuzzy Logic					(6 Hrs)
Introduction to Fuzzy Set: Fuzzy set theory, Fuzzy set Vs crisp set, Crisp relation & fuzzy relations, membership functions, Fuzzy Logic: Fuzzy Logic basics, Fuzzy Rules and Fuzzy Reasoning, Fuzzy inference systems: Fuzzification of input variables, defuzzification and fuzzy controllers.					
Unit III: Introduction to optimization Algorithm					(6 Hrs)
Introduction to Classical Methods & Linear Programming Problems Terminology, Design Variables, Constraints, Objective Function, Problem Formulation, Single Variable Optimization Problems, Multivariable and Constrained Optimization Techniques: Multi Variable and Constrained Optimization Technique, Optimality criteria , Direct search Method, Simplex search methods.					
Unit IV: Swarm Intelligence					(6Hrs)



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Intelligent Optimization Techniques, Swarm intelligence, Particle Swarm Optimization (PSO) Algorithm-Formulations, Pseudo-code, parameters, premature convergence, Real valued and binary PSO, Ant colony optimization (ACO)- Formulations, Pseudo-code. Applications of PSO and ACO. Grey Wolf Optimization - Formulations, Pseudo-code, Crow Search Algorithm.

List of Assignment

1. Implement fuzzy set operations using C/C++/Java language.
2. Implement fuzzy relational operations using C/C++/Java language
3. Write a program to solve a single objective function problem using Grey wolf optimization technique.
4. Write a program to solve a single objective function problem using CSA optimization technique.
5. Write a program to simulate the working nature of Ant Colony Optimization.
6. Solve some mathematical data based problems using the steps of PSO.

Text books:

1. S. N. Sivanandam and S. N. Deepa, Principles of soft computing – John Wiley & Sons, 2007.
2. Timothy J. Ross, Fuzzy Logic with engineering applications, John Wiley & Sons, 2016.

Reference Books:

1. N. K. Sinha and M. M. Gupta, Soft Computing & Intelligent Systems: Theory & Applications- Academic Press /Elsevier. 2009.
2. Simon Haykin, Neural Network- A Comprehensive Foundation- Prentice Hall International, Inc. 1998
3. R. Eberhart and Y. Shi, Computational Intelligence: Concepts to Implementation, Morgan Kaufman/Elsevier, 2007.
4. Driankov D., Hellendoorn H. and Reinfrank M., An Introduction to Fuzzy Control- Narosa Pub., 2001.

List of NPTEL / MOOC Courses:

1. **Introduction To Soft Computing By Prof. Debasis Samanta | IIT Kharagpur,**
https://onlinecourses.nptel.ac.in/noc22_cs54/preview
2. **NOC:Introduction to Soft Computing, IIT Kharagpur Prof. Debasis Samanta**
<https://nptel.ac.in/courses/106105173>
3. **NPTEL Video Course : NOC:Introduction to Soft Computing**
<http://www.digimat.in/nptel/courses/video/106105173/L01.html>
4. **Optimization Theory and Algorithms, By Prof. Uday Khankhoje | IIT Madras**
https://onlinecourses.nptel.ac.in/noc24_ee122/preview

CAUA32204B: CYBER SECURITY

Teaching Scheme	Examination Scheme				
Credits: 03	CIE	SCE	ESE	PR	Total
Lecture (L): 02 hrs / week					
Practical (P): 02 hr / week	20	20	40	20	100
Prerequisite:					
Course Objectives: 1. To understand cybercrime and ethical hacking. 2. To understand cryptography basics 3. To learn application of cybercrime tools. 4. To learn and apply penetration testing.					
Course Outcomes: After completion of the course, student will be able to 1. Remember the concept of cybercrime and ethical hacking. 2. Apply cryptography algorithms for secure communication. 3. Apply cybercrime tools for detecting and preventing vulnerabilities. 4. Apply penetration testing on applications.					
Unit I: Introduction to Cybercrime (6 Hrs)					
Cybercrime: Introduction, Cybercriminals, Classification of Cybercrimes, How Criminals Plan the Attacks, Cyberstalking, Credit Card Frauds in Mobile and Wireless Computing, Attacks on Mobile /Cell Phones. Ethical Hacking: Introduction, Need of Ethical Hacking, Types of Ethical Hacking, Prevention of Hacking. OWASP Top Ten					
Unit II: Cryptography Basics (6 Hrs)					
Security Attacks: Types of Attacks. Encryption Methods: Symmetric Encryption, Asymmetric Encryption. Cryptography: Hacking, Encryption, Decryption, Caser cipher, Monoalphabetic ciphers, Playfair Cipher, Single Columnar Transposition, Block Cipher Modes, Feistel Cipher, Data Encryption Standard, Advanced Encryption Standard.					
Unit III: Cybercrime Tools and Physical Security (6 Hrs)					
1) Vulnerability Scanners: Nessus, OpenVAS (Greenbone Vulnerability Manager), QualysGuard, 2) Web Application Vulnerability Scanners: Burp Suite, OWASP ZAP (Zed Attack Proxy), Acunetix. 3) Network Vulnerability Scanners: Nmap (Network Mapper, Nikto. 4) Log Manipulation Tools: Metasploit Framework, BleachBit, ExifTool					



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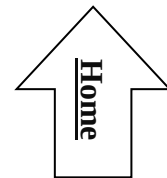
5) Anti-Forensic Tools: CCleaner, Chkrootkit Physical Security, Crime Prevention through Environmental Design (CPTED)	
Unit IV: Penetration Testing (6 Hrs)	
Business Continuity & Disaster Recovery Requirements, Phases of a Penetration Test Reconnaissance: HT Track, Google Directives, The Harvester, Who is, Netcraft, Host, Fierece and Other tools to Extract Information from DNS, Meta GooFi, Threat Agent (Attack of Drones), Sifting through the Intel to Find Attackable Targets.	
References: 1) Stallings, W. (2014). <i>Cryptography and network security: Principles and practice</i> (6th ed.). Pearson. 2)) Cryptography and Information Security, V. K. PACHGHARE, Nov 17, 2008, Computers, 384 pages.	
List of NPTEL Courses for SCE Assessment (Marks: 20) As part of SCE assessment, Student is required to successfully complete an NPTEL (National Programme on Technology Enhanced Learning) course. This activity is designed to enhance your subject knowledge, encourage self-learning, and expose you to expert insights on various topics. Below are the NPTEL Course names enrol for any one course and complete certification. 1) Cyber Security, Tools, Techniques and Counter Measures: https://onlinecourses.swayam2.ac.in/nou25_ge21/preview 2) Introduction to Cyber Security https://onlinecourses.swayam2.ac.in/nou25_cs04/preview 3) Information Security and Cyber Forensics https://onlinecourses.swayam2.ac.in/cec25_ge02/preview OR As part of SCE assessment, Student is required to prepare a PowerPoint presentation (PPT) on the research topic mentioned below (Choose any one topic). The purpose of this activity is to evaluate understanding of the topic, ability to convey information effectively, and presentation skills. (Note: Individual Assessment) Topics of SCE presentation based on various Research Areas: 1) Security in IoT and Cloud Computing. 2) AI in Cyber Security. 3) IoT and Mobile Security. 4) Defensive and Offensive Security. 5) Cyber Forensics. 6) Cryptography. 7) Threat Intelligence and Incident Response. 8) Application, Endpoint and Data Security. 9) Network Security. Drones Security	
List of Assignments:	
1	Decoding Email Headers: A Practical Approach to Identifying and Mitigating Cyber Threats. Objective:



	To understand and analyse email headers to detect signs of phishing, spoofing, or other malicious activities. This assignment aims to equip students with practical skills for email forensics and cybersecurity analysis.
2	Exploring the Role and Implementation of SSL Certificates in Secure Communication. Objective: To understand the significance, creation, and practical usage of SSL certificates in ensuring secure web communications. This assignment will emphasize both theoretical knowledge and hands-on experience with SSL certificates.
3	Unveiling the Power of Google Hacking: Techniques, Risks, and Ethical Implications Objective: To understand and utilize Google Hacking techniques for ethical purposes, such as vulnerability assessment, information gathering, and cybersecurity awareness. This assignment aims to teach students the responsible use of publicly accessible data and highlight the potential risks posed by improper use.
4	Develop the program to implement DES algorithm for encryption and decryption. Assume suitable key. Objective: The Data Encryption Standard (DES) is a symmetric-key block cipher designed to encrypt and decrypt data in a secure and efficient manner. Its primary objective is to provide a method of securing electronic communications and data storage.
5	Develop the program to implement RSA algorithm for encryption and decryption. Assume suitable Private and Public Keys. Objective: RSA provides a way to encrypt sensitive data so that it can only be decrypted by the intended recipient. By using a pair of public and private keys, RSA ensures that even if the public key is exposed, the data remains confidential as only the corresponding private key can decrypt it.
6	Develop the program of Euclid's algorithm to find the GCD of two integers. Objective: Euclid's Algorithm is a classical method for finding the Greatest Common Divisor (GCD) or Greatest Common Factor (GCF) of two integers. The objective of Euclid's Algorithm is to provide an efficient and systematic way to compute the GCD, which is the largest positive integer that divides both numbers without leaving a remainder.
7	Investigate Vulnerability Scanners: • Explore and analyze the following vulnerability scanners: ◦ Nessus ◦ OpenVAS (Greenbone Vulnerability Manager) ◦ QualysGuard • Identify the types of vulnerabilities each scanner targets (e.g., network vulnerabilities, web application vulnerabilities, etc.). • Compare the features of each scanner, including detection accuracy, ease of use, and integration with other security tools. Evaluate the strengths and limitations of each tool in different security scenarios.
8	Web Application Vulnerability Scanners: • Research the following web application scanners: ◦ Burp Suite ◦ OWASP ZAP (Zed Attack Proxy)



	○ Acunetix • Analyse their capabilities in detecting vulnerabilities such as SQL injection, XSS (Cross-Site Scripting), and CSRF (Cross-Site Request Forgery). Discuss how these scanners fit into a penetration testing workflow and the benefits of automating web application vulnerability scans.
9	Network Vulnerability Scanners: • Examine Nmap (Network Mapper) and Nikto as network vulnerability scanners. • Investigate how Nmap is used for network discovery and vulnerability scanning. • Evaluate how Nikto is used for detecting web server vulnerabilities. Provide use cases where these scanners are most effective and discuss their limitations in identifying network vulnerabilities.
10	Log Manipulation Tools: • Investigate the role of log manipulation tools in cybersecurity investigations and system administration. ○ Metasploit Framework ○ BleachBit ○ ExifTool • Explore how these tools are used for log analysis, file manipulation, and removing metadata to conceal activities during penetration testing or forensic investigations. Compare the features and functionalities of each tool in terms of user-friendliness, log manipulation capabilities, and integration with other tools.
11	Anti-Forensic Tools: • Study the anti-forensic tools listed below and their role in covering up traces of digital footprints. ○ CCleaner ○ Chkrootkit • Explore the functionality of each tool in deleting files, cleaning system traces, and scanning for rootkits. Discuss ethical concerns related to the use of anti-forensic tools and their potential misuse in criminal activities or to circumvent digital investigations.



AIUA32204D: AIML WITH MAINFRAME TECHNOLOGY

Teaching Scheme	Examination Scheme				
Credits: 04	CIE	SCE	Project	PR	Total
Lecture (L): 03 hrs / week					
Practical (P): 02 hr / week	20	20	40	20	100
Prerequisite: Nil					
Course Objectives:- 1. To introduce IBM Z Mainframe Technology. 2. To learn industry-critical enterprise computing skills 3. To introduce IBM Mainframe Operating System/OS. 4. To learn /OS Access and Interaction Basics 5. To study the Operating System Environment for computing. 6. To learn /OS REXX, COBOL Programming.					
Course Outcome:- After completion of the course, student will be able to: 1: Understanding of IBM Z Mainframe Technology. 2: Apply the industry-critical enterprise computing skills. 3: Apply IBM Mainframe Operating Systemz/OS for Large Enterprise. 4: Apply z/O s Access and Interaction Basics. 5: Design and develop a program for Job Control Language using z/OS. 6: Design and develop a program for REXX, COBOL using z/OS.					
Unit I: Introduction to IBM Mainframe Technology					
Introducing IBM Z Hardware, IBM z16, IBM z16 single frame, IBM z16 rack mount, IBM z15 T01 and T02 machines, Applications of IBM Z, Modern Mainframe innovations: z15 and z16 systems, Containerization and Kubernetes on Mainframes, Mainframe Hardware components such as Central Processing Unit (CPU), Memory and Storage Systems, Input/Output Devices, Case studies of Mainframe implementations.					
Unit II: Learn Industry-Critical Enterprise Computing Skills					
IBM Z Mainframe Skills Depot: Mainframe System Administrator, Mainframe Application Developer/Programmer, Mainframe Modernization Architect, Database Administrator (DBA), Storage Administrator, System Programmer (SysProg), Network Administrator, Mainframe DevOps Engineer, Cybersecurity Engineers, to name a few job roles in the vast and amazing world of the IBM Z Mainframes.					
Unit III: z/OS Introduction					
z/OS: Essential Technology of Large Enterprises, Introducing IBM Mainframe Operating Systems, z/OS-IBM Z Flagship Operating System, z/OS Professional Documentation Architecture of IBM Z systems, networking, and security aspects					
Unit IV: z/OS Access and Interaction Basics					
Time Sharing Option, Interactive System Productivity Facility (ISPF) Editor, System Display and Search Facility (SDSF), TN3270, SSH, FTP.					



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Datasets and Data Access Methods: Dataset Names and Types, Sequential Dataset, Partitioned Datasets, VSAM - Virtual Storage Access Method, Disk Storage, Volume Labels, VTOC, Extents, System Utilities and Tools, Dataset management utilities (IEBGENER, IDCAMS, etc.), DFSORT basic and examples.

Unit V: Operating System Environment

Unix System Services (USS), JCL - Job Control Language, JES - Job Entry Subsystem, System Catalog, Components, Messages, System Log, Virtual Storage and Address Spaces, System and User Address Space Management, Key Controlled Protection for Address Spaces, Security, RACF, Communications Server, z/OS Operator Commands, z/OS Initial Load Program Process, Virtualization and Cloud computing.

Unit VI: Programming

z/OS REXX programming, Learning COBOL programming with VS code, z/OS Unix System Services (USS), JCL (Job Control Language), Interactive System Productivity Facility (ISPF), Java on z/OS, Virtual Storage Access Method (VSAM), Ansible Automation, creating fully automated CI/CD pipelines involving components residing on z/OS, Zowe, Big Data and Analytics: Hadoop on Mainframes, Machine Learning Integrations. Mainframe Application Lifecycle Management. Agile project Methodology.

e-Resources:

1. https://ibmzxplere.influitive.com/users/sign_in
2. <https://www.ibm.com/z/education>

MOOC Courses:

1. <https://www.ibm.com/training/>
2. <https://learn.ibm.com/>
3. <https://www.ibm.com/z/education>
4. <https://www.redbooks.ibm.com/redbooks.nsf/redbookabstracts/crse0304.html>

Job Control Language (JCL) Assignments:

- | | |
|----------|--|
| 1 | Data set operations (Create, Delete, Rename, Copy, Catalog, Uncatalog, etc.) |
| 2 | Submit Job, Redirect Output, Check Job Status and Output |
| 3 | Condition code handling, JCL overrides |

COBOL Programming Assignments:

- | | |
|----------|--|
| 4 | Control Structures: If-Else Condition and Simple Loop(PERFORM UNTIL) |
| 5 | File Handling: Sequential and Indexed (VSAM) |
| 6 | Perform varying, Error handling, etc. |

Sample Project List:(<https://openmainframeproject.org/>)



Bansilal Ramnath Agarwal Charitable Trust's

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- | | |
|--|--|
| | <ol style="list-style-type: none">1. To Write a COBOL application program that reads input from three files, merge their contents, and writes to an output file containing one sorted list.2. Demonstrate file-handling capabilities using a REXX program.3. To build an end-to-end CI/CD DevOps Pipeline for the mainframe environment. |
|--|--|

**IOEUA32205C: EXPLAINABLE ARTIFICIAL INTELLIGENCE (XAI) FOR ENGINEERING APPLICATIONS**

Teaching Scheme	Examination Scheme				
Credits: 03 Lecture (L): 03 hrs / week	CIE	SCE	ESE	OR	Total
	20	20	40	20	100
Prerequisite: Fundamentals of Probability & Statistics, Machine Learning and Deep Learning basics, Python for Data Science					
Course objectives: 1. Making students familiarize with the need of XAI for engineering applications and its central concepts 2. Making students understand with the mathematical concepts like ensemble models and non- linear models to analyse the problems 3. Providing tools and techniques of XAI for design and building solutions					
Course Outcomes: After completion of the course, student will be able to 1. Learn the fundamental concepts of XAI and its use to build various use cases in engineering domain 2. Compare merits and demerits of linear and non-linear model in problem analysis 3. Provide knowledge about using ensemble learning and contrastive explanations and LRP for machine learning 4. Performs parametric evaluation of AI-based and XAI-based solutions 5. Apply the knowledge for drafting clear requirements to build end-to-end XAI solution 6. Learn and apply knowledge of XAI and tools for application and protocol development in engineering applications					
Unit I: Introduction to Explainable Artificial Intelligence					
Artificial Intelligence, Need for XAI, Explainability vs. Interpretability Explainability Types: Intrinsic explanation, Post-hoc explanation, Model specific, Model agnostic, Local interpretation, Global interpretation, Sublocal interpretation, Textual explanations, Visual explanations Tools for Model Explainability: SHAP, LIME, ELI5, Skater, Skope_rules Evolution of XAI, Biasness, and Reliability, Challenges to achieve explainable AI and design issues Case Studies: Fraud Detection, Online Recommendations, Credit and Loan Decision Making					
Unit II: Explainability for Linear Models					
Linear Models, Linear Regression VIF and the Problems It Can Generate : Final Model, Model Explainability Trust in ML Model: SHAP - Local Explanation and Individual Predictions in a ML Model, Global Explanation and Overall Predictions in ML Model, LIME Explanation and ML Model, Skater Explanation and ML Model, ELI5 Explanation and ML Model, Logistic Regression: Interpretation, LIME Inference Case Studies: Linear Regression					
Unit III: Explainability for Non-Linear Models					



Non-Linear Models Decision Tree Explanation, Data Preparation for the Decision Tree Model Creating the Model , Decision Tree — SHAP, Partial Dependency Plot, PDP Using Scikit-Learn, Non-Linear Model Explanation — LIME, Non-Linear Explanation — Skope-Rules Case Studies: Comparison of Husky Dog and Wolf
Unit IV: Explainability for Ensemble Models
Ensemble Models: Types of Ensemble Models Why Ensemble Models?, Using SHAP for Ensemble Models, Using the Interpret Explaining, Boosting Model, Ensemble Classification Model: SHAP, Using SHAP to Explain Categorical Boosting Models, Using SHAP Multiclass Categorical Boosting Model, Using SHAP for Light GBM Model Explanation Case Studies: Model Interpretability
Unit V: Counterfactual Explanations for XAI Models
AI Model Fairness Using a What-If Scenario: What Is the WIT (Google Tool)?, Evaluation Metric. Counterfactual Explanations for XAI Models: What Are CFEs?, Implementation of CFEs, CFEs Using Alibi, Counterfactual for Regression Tasks. Case Studies: Causability Algorithms and Applications
Unit VI: Contrastive Explanations and LRP for Machine Learning
What Is CE for ML?, CEM Using Alibi, Comparison of an Original Image vs. an Autoencoder. Generated Image, CEM for Tabular Data Explanations. Layer wise relevance propagation (LRP): Introduction, Working Principle, Mathematical Modeling. Case Studies: Pertinent Negatives, Explanation based on missing
Reference Books: Hands-On Explainable AI (XAI) with Python: Interpret, visualize, explain, and integrate reliable AI for fair, secure, and trustworthy AI apps by Denis Rothman

**IOEUA32205B: ENGINEERING ECONOMICS AND FINTECH**

Teaching Scheme	Examination Scheme				
Credits: 03 Lecture (L): 03 hrs / week	CIE	SCE	ESE	OR	Total
	20	20	40	20	100
Prerequisite: NIL					
Course objectives: 1. To learn about the economic development and related issues 2. To explain the Indian banking structure and terms like GDP, inflation 3. To introduce Cash Flow analysis and Taxes 4. To introduce FinTech and it's sub sectors 5. To explain the classification of various models of FinTech 6. To describe the innovation in FinTech					
Course Outcomes: After completion of the course, student will be able to 1. Understand the fundamental concepts in engineering economics 2. Illustrate the terms like GDP, inflation, and Indian banking structure 3. Analyze and Calculate cash flow analysis and taxes 4. Understand what FinTech is and the sub sectors that comprise it 5. Classify various models of the Fintech 6. Illustrate various innovations done using latest technology trends in FinTech					
Unit I: Introduction to Economics					
Introduction to Economics- Flow in an economy, Law of supply and demand, Concept of Engineering Economics – Engineering efficiency, Economic efficiency, Scope of engineering economics – Element of costs, Marginal cost, Marginal Revenue, Sunk cost, Opportunity cost.					
Unit II: Banking and Trades					
Indian Economics – basic features, natural resources, population size and composition, national income concepts, microeconomics of India, Indian Banking, Role of Reserve bank of India, International Economy, inflation, GDP, Financial Analysis, Ratios, financial Institutions, Finance Commissions.					
Unit III: Understanding Cash Flow analysis					
Budget Analysis, Break-even analysis, Elementary economic Analysis – Material selection for product, Accounting for Depreciation, Project Cash-Flow Analysis, Understanding Financial Statements.					
Unit IV: Introduction to FinTech					
Introduction, Financial Services and Fintech: Introduction, Changing Environment, Customer Centricity, Digital Transformation, Definition of Fintech, History of Fintech, Fintech stages, An Overview of Fintech Initiatives Around the World, Ecosystems, Downsides of Disruptive Fintech					
Unit V: Model and Classifications					
Introduction, Classification, Five Ws and one H : 1. Why a fintech initiative was born? 2. For whom was it born? 3. Which are the services it aims to provide? 4. Where does it aim to perform its business? 5.					



When does it aim to operate, within the framework of the financial cycle? 6. How is fintech working? The organization and its elements, The V4 business model framework, A Business Model, A Business Model for Fintech. Business Model Canvas (BMC) for FinTech.

Unit VI: FinTech Innovation

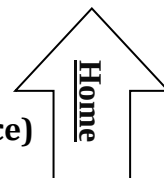
Innovation and Fintech, Digital Transformation and Fintech, A model for an integrated innovation strategy, Types of Innovation: Product (or services), Process, Organization, Examples of Innovation, Process Innovation: Big Data Analytics, Value Creation from Big Data Analytics, Kreditech's self-learning algorithm, Internet of Things, Blockchain Technology, Organizational Innovation: Social Networks

Text Books:-

1. Nicoletti, The Future of FinTech, 1st ed. Palgrave Macmillan, 2017
2. Krugman, International Economics, Pearson Education
3. Thursen Gerald, Engineering Economics, Prentice Hall

Reference Books:

1. Accenture. (2015). The future of FinTech and banking: Digitally disrupted or reimaged?
2. Accenture Research, 1–12
3. Dietz M., Khanna S., Olanrewaju T., and Rajgopal K. (2015). Cutting through the fintech noise: Markers of success, imperatives for banks. Practice, G. B. (Ed.), 1–18. McKinsey and Company.
4. Retrieved from <http://www.mckinsey.com/industries/financial-services/our-insights/cutting-through-the-noise-round-financial-technology>.
5. "What is FinTech and why does it matter to all entrepreneurs?". *Hot Topics. July 2014*.retrieved December 9, 2014.



AIUA32204E: MACHINE LEARNING OPERATIONS

Teaching Scheme	Examination Scheme									
Credits: 03 Lecture (L): 02 hrs / week Practical(P): 02 hrs / week	HA	TW	SCE	PPT	GD	CIE	ESE	PR	OR	Total
		20	20					40	20	100
Prerequisites: Cloud Deployment, Modern Integration with cloud, AWS to deploy, experience with python programming, Flask k not to look at local deployment, Familiarity with data handling and processing, A basic understanding of machine learning concepts and models.										
Course Objective(s): - To provide students with a comprehensive understanding of the fundamental principles, concepts, and techniques of machine learning. - To enable students to design and implement various machine learning algorithms and models. - Data Handling and Preprocessing: To equip students with skills to preprocess, analyze, and visualize data for effective machine learning applications. - Model Evaluation: To teach students how to evaluate the performance of machine learning models using appropriate metrics and validation techniques. - Application of Machine Learning: To foster the ability to apply machine learning methods to solve real-world problems in various domains.										
Course Outcomes: - Understanding Machine Learning Concepts: Students will be able to explain the fundamental concepts and principles of machine learning - Algorithm Implementation: Students will be proficient in implementing various machine learning algorithms - Data Preprocessing Skills: Students will be able to preprocess and transform raw data into a suitable format for machine learning models, including handling missing values, scaling features, and encoding categorical variables. - Model Building and Training: Students will be capable of building, training, and optimizing machine learning models using appropriate algorithms and frameworks. - Performance Evaluation: Students will be skilled in evaluating the performance of machine learning models using. - Application Development: Students will be able to apply machine learning techniques to develop solutions for practical problems.										
Contents										
Introduction to MLOps (4 Hrs)										
Overview of MLOps and its importance, Difference between MLOps, DevOps and DataOps, 3. Key concepts and terminologies in MLOps.										



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Module 1: Machine Learning Lifecycle	(4 Hrs)
1. Understanding the ML lifecycle stages 2. Problem definition 3. Data pre-processing and feature engineering 4. Model development, training, and evaluation 5. Model deployment strategies 6. Model monitoring, maintenance, and continuous improvement	
Module 2: MLOps Tools and Technologies	(4 Hrs)
1. Overview of MLOps tools and platforms 2. Introduction to version control systems (e.g., Git) 3. Data versioning tools (e.g., DVC) 4. Model packaging – Fast API, Pickle files 5. Experiment tracking and management (e.g., Kubeflow) 6. Model serving frameworks (e.g., Kserve) 7. Containerization with Docker 8. Orchestration with Kubernetes	
Module 3: MLOps Infrastructure	(4 Hrs)
1. Setting up scalable ML infrastructure 2. Cloud computing services (AWS) 3. Infrastructure as Code (IaC) using tools like Terraform 4. Continuous Integration and Continuous Deployment (CI/CD) for ML 5. End-to-end automation pipelines	
Module 4: Data and Model Management	(4 Hrs)
1. Data storage and pipelines 2. Feature stores and their role in MLOps 3. Model registry and model store concepts – AWS Sagemaker 4. Model governance – Model Explainability	
Module 5: Deployment Strategies	(4 Hrs)
1. Batch inference vs. real-time inference 2. Canary releases and A/B testing for models 3. Multi armed Bandits	
Module 6: Monitoring and Operations	(3 Hrs)
1. Logging, monitoring, and alerting for ML systems 2. Monitoring and Tracing 3. Model performance metrics and evaluation 4. Incident management and troubleshooting	
Module 7: Security, Privacy, and Compliance	(3 Hrs)
1. Security best practices for MLOps	



2. Data privacy and handling sensitive information

3. Ethical considerations in ML operations

Module 8: Best Practices and Case Studies

(3 Hrs)

1. Best practices for implementing MLOps in organizations

2. Case studies of MLOps in the Agriculture sector

Module 9: Hands-On Projects

(3 Hrs)

1. Setting up a complete MLOps pipeline

2. Deploying a machine learning model to production

3. Monitoring and maintaining a deployed model

4. Implementing CI/CD for a machine learning project

CAUA32205: PROJECT - II

Teaching Scheme	Examination Scheme					
Credits: 02 Practical (P): 04 hrs / week	CIE	ISE	SCE	OR	TW	Total
	--	--	--	--	50	50
Prerequisites:						
Course Objective(s): - To apply the learnt skill sets required to solve an engineering related problem. - To expose students to product development cycle using industrial experience, use of state of art technologies. - To provide the students with the opportunity to design or conduct independent research related to IT courses. - To encourage and expose students for participation in research paper presentation activities and IPR. - To learn the importance of teamwork, documentation and ethics.						
Course Outcomes: Upon completion of course, students will be able to: - Identify the problem that needs technological solution and define scope. - Demonstrate a sound literature review and design the solution. - Implement the solution and test by applying knowledge and various tools. - Track the progress of the project with the help of a project management tool. - Perform effectively as an individual and a team member in multi-disciplinary teams. - Communicate with engineers and the community at large in written and oral forms.						
Contents						
Unit I: - Title and scope finalization (6 Hrs)						
Problem statement: a concise statement of the problems which initiate the industrial problem or research questions or design ideas. Answering questions: What is the issue that we want to address (problem or question)? Why need to address the issues? How can the project solve the issues?, d. Who will benefit from the project? Setting a clear goal of what needs to be accomplished by doing the work. Students should						
Objectives: only state the technical objective of the project (e.g. to evaluate the performance of the design, to test a hypothesis, to study the relationship between variable x and variable y etc.). Use measurable action verbs when defining an objective (e.g. define, design, identify, describe, analyze, evaluate etc).						
Scope: Scope sets a clear boundary (time, geography, environment, function etc.) of our work to provide a common understanding of the project among stakeholders (in FYP the stakeholders are students, lecturer, panels etc.). Scope makes our project achievable and realistic by defining the limits and constraints of the study.						
Formal technical review-0 (FTR-0)						



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Unit II: Literature review and design (6 Hrs)

Literature review: To summarize and synthesize the ideas of others (Similar systems and research work) Discuss the techniques or different design decisions; make comparisons and give technical comments; Summarize the comparison with tables or graphs to clarify the differences.

Design: SRS; Use of the designing tools; discussion on the tradeoff of a particular design decision; Selection of Technology and Tools, UI/UX design, database design, design of test cases and performance metrics Installations; Use of various tools

Formal technical review-1 (FTR-1)

Unit III: Implementation and Testing (6 Hrs)

Implementation of algorithms, databases and modules as per design; Testing: Analysis of results and performance; comparative analysis (comparison with existing/ known algorithms/ systems) and validation of results; discussions and conclusions.

Formal technical review-2 (FTR-2)

Unit IV: Report writing and Project outcome generation (6 Hrs)

Documentation and presentation: Project report writing and demo video creation. The student shall prepare and submit the report of Project work in standard format for satisfactory completion of the work that is duly certified by the concerned guide and head of the Department/Institute.

Project outcome: Submit the research paper based on the work done to a reputed conference/ journal or IPR or participation in hackathon or participation in project competition as per the suggestions of the guide. For industry sponsored projects, the work must be submitted/ deployed as per industry requirement and obtain the completion certificate from the industry.

Project Oral exam: The assessment will be broadly based on work undergone, content delivery, presentation skills, documentation, question-answers and report.